

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 10       | (a) | (i)  |  | temporary hard water (1)<br><br>award (1) for any of following <ul style="list-style-type: none"><li>same volume of soap solution needed (to produce a lather) after boiling as sample <b>C</b> / soft water</li><li>same volume of soap solution needed (to produce a lather) after boiling as after adding washing soda</li><li>it is soft after boiling / hardness is removed by boiling</li></ul> neutral answers -<br>less soap needed after boiling<br>hardness decreased after boiling |                 |     | 2   | 2     |       | 2    |
|          |     | (ii) |  | <b>D</b> (1)<br><br>award (1) for either of following <ul style="list-style-type: none"><li>less soap solution needed (to produce a lather) after boiling but still more than needed for sample <b>C</b> / soft water</li><li>boiling did not remove all the hardness / partly softened by boiling</li><li>less soap needed after boiling and less again after washing soda</li></ul>                                                                                                         |                 |     | 2   | 2     |       | 2    |

| Question |     |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |     |     |       |       |      |
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|          |     |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (b) |     |  | <p>award (1) for similarity in results e.g.</p> <ul style="list-style-type: none"> <li>(both groups can draw the) same conclusions for samples <b>A</b>, <b>C</b> and <b>D</b></li> <li>volumes in the second experiment are approximately three quarters the values in the first experiment for samples <b>A</b>, <b>C</b> and <b>D</b></li> <li>results for samples <b>A</b>, <b>C</b> and <b>D</b> are reproducible</li> </ul> <p>award (1) for different conclusions drawn for sample <b>B</b> e.g.</p> <ul style="list-style-type: none"> <li>first group found sample <b>B</b> to be permanently hard water but the second group found that it contains some temporary hardness and some permanent hardness</li> </ul> |                 |     | 2   | 2     |       | 2    |
|          | (c) |     |  | <p>calcium / magnesium <b>ions</b> <u>swap places with</u> / replaced by / exchanged for sodium <b>ions</b> (2)</p> <p>there must be one reference to <u>ions</u></p> <p>if not complete answer award (1) for either of following<br/>calcium / magnesium <b>ions</b> removed / replaced<br/>sodium <b>ions</b> added</p> <p>if no mention of <b>ions</b> anywhere award (1) for<br/>calcium / magnesium swaps places with / replaced by / exchanged for sodium</p> <p>do not accept 'displace'</p>                                                                                                                                                                                                                          | 2               |     |     | 2     |       |      |
|          | (d) | (i) |  | <p><math>3.2 \times 10^{-3} \text{ mol}</math> (2)</p> <p>award (1) for 0.0032</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 | 2   |     | 2     | 2     |      |

| Question |  |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                    | Marks available |     |     |       |       |      |
|----------|--|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |      |  |                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (ii) |  | moles $\text{MgCl}_2 = \frac{0.384}{95} = 4.04 \times 10^{-3} \text{ mol}$<br><br>more <b>Mg<sup>2+</sup> ions</b> in second sample / magnesium chloride sample<br>therefore it has more hardness (1)<br>accept more moles of magnesium chloride than magnesium sulfate<br>(in the same mass because $M_r$ of magnesium chloride is smaller)<br><br>ecf possible from part (d) (i) |                 |     | 2   | 2     | 1     |      |
|          |  |      |  | Question 10 total                                                                                                                                                                                                                                                                                                                                                                  | 2               | 2   | 8   | 12    | 3     | 6    |